Topology
Open and Closed Maps

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Contents

1 Introduction

In our previous lectures, we explored the concept of **continuous maps**, where the inverse image of every open set in the codomain is an open set in the domain. Today, we will shift our focus to the forward direction and study two related but distinct types of maps: **Open Maps** and **Closed Maps**. These concepts are crucial for understanding how topological properties are preserved under different kinds of functions.

- **Open Map:** Preserves the "openness" of sets when mapping forward.
- **Closed Map:** Preserves the "closedness" of sets when mapping forward.

Reference

This lecture is based on concepts from "Topology" by James R. Munkres.

Big Picture

Think of a function as a "machine" that transforms one space into another. We have three ways to measure how "nice" this machine is:

- **Continuous:** Open sets *pull back* to open sets (looking *backward*).
- **Open map:** Open sets *push forward* to open sets (looking *forward*).
- **Closed map:** Closed sets *push forward* to closed sets (looking *forward*).

These are independent properties — a map can have any combination of them.

2 Open Map

2.1 Definition

Intuition: Open Map

An open map "respects openness in the forward direction." If you start with an open neighbourhood around a point and push it through f, the image is still an open neighbourhood around the image point. Informally: *open sets are not "squashed" or "collapsed" in a way that destroys openness.*

Definition: Open Map

A map (or function) $f : X \to Y$ from a topological space (X, τ_X) to another topological space (Y, τ_Y) is called an **open map** if for every open set A in X (i.e., $A \in \tau_X$), its image $f(A)$ is an open set in Y (i.e., $f(A) \in \tau_Y$).

In simple terms: An open map sends open sets to open sets.

2.2 Visualizing Open Maps with a Diagram

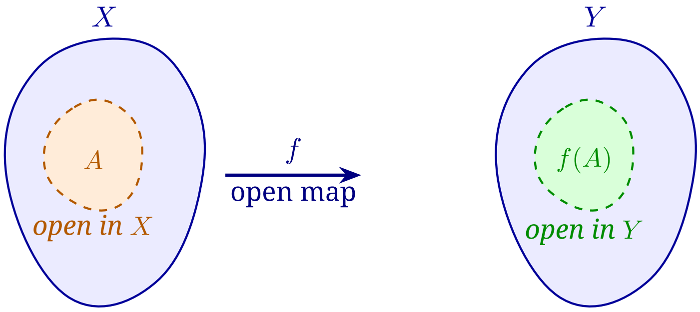


Figure 1: An **open map** sends every open set $A \subseteq X$ to an open set $f(A) \subseteq Y$.

2.3 Example of an Open Map (Original)

Example 1

Let's consider two topological spaces (X, τ_X) and (Y, τ_Y) and a map f between them.

Step 1: Define Sets and Topologies:

- Domain set: $X = \{1, 2, 3\}$
- Codomain set: $Y = \{a, b, c\}$
- Topology on X: $\tau_X = \{\emptyset, X, \{1\}\}$
- Topology on Y: $\tau_Y = \{\emptyset, Y, \{a\}, \{b\}, \{a, b\}\}$

Step 2: Define the Map $f : X \to Y$:

$$f(1) = b, \quad f(2) = c, \quad f(3) = a$$

Step 3: Check the Condition for an Open Map:
We must find the image of *every* open set in X and check if that image is open in Y.

- **Open Set 1:** $\emptyset$
  - The image is $f(\emptyset) = \emptyset$.
  - Is $\emptyset$ an open set in Y (i.e., is $\emptyset \in \tau_Y$)? **Yes.**
- **Open Set 2:** $X = \{1, 2, 3\}$
  - The image is $f(X) = f(\{1, 2, 3\}) = \{f(1), f(2), f(3)\} = \{b, c, a\} = Y$.
  - Is Y an open set in Y (i.e., is $Y \in \tau_Y$)? **Yes.**
- **Open Set 3:** $\{1\}$
  - The image is $f(\{1\}) = \{f(1)\} = \{b\}$.
  - Is $\{b\}$ an open set in Y (i.e., is $\{b\} \in \tau_Y$)? **Yes.**

Step 4: Conclusion:
Since the image of every open set in X is an open set in Y, the map f **is an open map**.

2.4 Additional Examples of Open Maps

Example 2.1 (A Map that is **NOT** an Open Map)**.** *Let* $X = \{1,2,3\}$*,* $Y = \{a,b,c\}$*, with topologies:*

$$\tau_X = \{\emptyset,\ X,\ \{1\}\}, \qquad \tau_Y = \{\emptyset,\ Y,\ \{a,b\}\}.$$

Define $f : X \to Y$ *by* $f(1) = a,\ f(2) = b,\ f(3) = c$*.*

Check each open set in X***:***

- $f(\emptyset) = \emptyset \in \tau_Y$. ✓
- $f(X) = Y \in \tau_Y$. ✓
- $f(\{1\}) = \{a\}$. *Is* $\{a\} \in \tau_Y$*?* ***No!*** τ_Y *contains* $\{a,b\}$ *but not* $\{a\}$ *alone.*

Conclusion: *The image of the open set* $\{1\}$ *is* $\{a\}$*, which is* not *open in* Y*. Therefore,* f *is* ***not an open map****.*

Key Lesson: It only takes one *open set whose image is not open to disqualify a map from being open.*

Example 2.2 (Identity Map is Always Open)**.** *Let* (X,τ) *be* any *topological space. Define* $\text{id} : X \to X$ *by* $\text{id}(x) = x$*.*

For any open set $U \in \tau$*, we have* $\text{id}(U) = U \in \tau$*.*

Conclusion: *The identity map on any topological space is always an open map (and in fact also a closed map and a homeomorphism).*

Example 2.3 (Open Map with Indiscrete Codomain)**.** *Let* $X = \{a,b\}$ *with discrete topology* $\tau_X = \{\emptyset,\{a\},\{b\},X\}$*, and let* $Y = \{p,q\}$ *with the indiscrete topology* $\tau_Y = \{\emptyset, Y\}$*.*

Define $f : X \to Y$ *by* $f(a) = p,\ f(b) = q$*.*

Check each open set in X***:***

- $f(\emptyset) = \emptyset \in \tau_Y$. ✓
- $f(\{a\}) = \{p\}$. *Is* $\{p\} \in \tau_Y$*?* ***No!*** *The only open sets in* Y *are* $\emptyset$ *and* Y*.*

Conclusion: f *is* ***not an open map****. When the codomain carries very few open sets (indiscrete), it is hard for a non-constant map to be open.*

Contrast: Any map into *an indiscrete space with* indiscrete *domain is trivially open, since the only open sets in* X *are* $\emptyset$ *and* X*, whose images* $\emptyset$ *and* Y *are always open.*

⚠ Common Mistakes: Open Maps

1. **Confusing direction.** Continuity checks $f^{-1}(V)$ for open $V \subseteq Y$. Open maps check $f(U)$ for open $U \subseteq X$. These are *opposite* directions.
2. **Assuming continuity implies openness.** A continuous map need not be open, and an open map need not be continuous. They are genuinely independent properties.

2.5 A Useful Proposition

Proposition 2.1. *Let* $f : X \to Y$ *be a bijection. Then* f *is a homeomorphism if and only if* f *is continuous* and *open.*

Proof. (⇒) If f is a homeomorphism then f and f^{-1} are both continuous. For any open $U \subseteq X$, continuity of f^{-1} gives $(f^{-1})^{-1}(U) = f(U)$ is open in Y. So f is an open map.

(⇐) Suppose f is a continuous bijection and an open map. We must show f^{-1} is continuous, i.e., for every open $V \subseteq Y$, $(f^{-1})^{-1}(V) = f(V)$ is open in Y. But this is exactly the open map condition. □ □

Section 2 Summary: Open Maps

- **Definition:** $f : X \to Y$ is an **open map** if $f(U) \in \tau_Y$ for every $U \in \tau_X$.
- **Key check:** Compute the image of *every* open set in X and verify it is open in Y. One failure is enough to disqualify.
- **Trivial cases:** $\emptyset$ and X always satisfy the condition. Focus on the non-trivial open sets.
- **Independence:** Openness and continuity are independent. A map can be open but not continuous, continuous but not open, both, or neither.
- **Bijection:** A bijection f is a homeomorphism $\iff$ f is continuous and open.

3 Closed Map

3.1 Definition

Intuition: Closed Map

A closed map is the "companion" to an open map, but for closed sets. Closed sets are complements of open sets — they contain all their limit points. A closed map guarantees that if you start with a set that "contains its boundary," the image also "contains its boundary" in Y. Neither continuity nor openness implies this directly.

Definition: Closed Map

A map $f : X \to Y$ is called a **closed map** if for every closed set B in X, its image $f(B)$ is a closed set in Y.

In simple terms: A closed map sends closed sets to closed sets.

Key Concept: Finding Closed Sets

To find the closed sets of a topological space, we take the complements of all its open sets relative to the universal set.

3.2 Visualizing Closed Maps

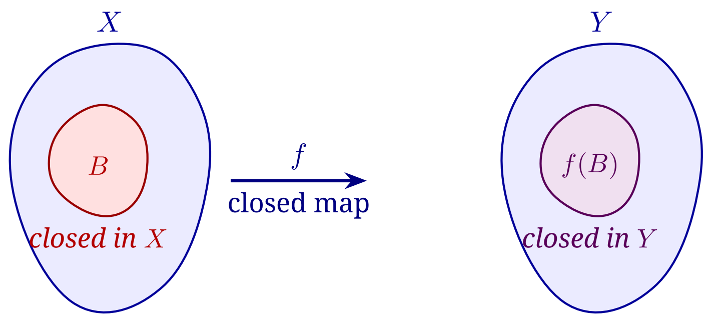


Figure 2: A **closed map** sends every closed set $B \subseteq X$ to a closed set $f(B) \subseteq Y$.

3.3 Example of a Closed Map (Original)

Example 2

Let's use a new example to illustrate the concept.

Step 1: Define Sets and Topologies:

- $X = \{1, 2\}$
- $Y = \{a, b, c\}$
- $\tau_X = \{\emptyset, X, \{1\}\}$
- $\tau_Y = \{\emptyset, Y, \{a\}, \{a, b\}\}$

Step 2: Find the Closed Sets in X and Y:

- **Closed Sets in X** (Complements of sets in τ_X): $\{X, \emptyset, \{2\}\}$
- **Closed Sets in Y** (Complements of sets in τ_Y): $\{Y, \emptyset, \{b, c\}, \{c\}\}$

Step 3: Define the Map $f : X \to Y$:

$$f(1) = b, \quad f(2) = c$$

Step 4: Check the Condition for a Closed Map:

- **Closed Set 1:** $X = \{1, 2\}$
  - The image is $f(X) = f(\{1, 2\}) = \{f(1), f(2)\} = \{b, c\}$.
  - Is $\{b, c\}$ a closed set in Y? **Yes.**
- **Closed Set 2:** $\emptyset$
  - The image is $f(\emptyset) = \emptyset$.
  - Is $\emptyset$ a closed set in Y? **Yes.**
- **Closed Set 3:** $\{2\}$
  - The image is $f(\{2\}) = \{f(2)\} = \{c\}$.
  - Is $\{c\}$ a closed set in Y? **Yes.**

Step 5: Conclusion:
Since the image of every closed set in X is a closed set in Y, this map f **is a closed map**.

3.4 Additional Examples of Closed Maps

Example 3.1 (A Map that is **NOT** a Closed Map). *Let $X = \{1,2,3\}$, $Y = \{a,b,c\}$, with:*

$$\tau_X = \{\emptyset,\ X,\ \{1\}\}, \qquad \tau_Y = \{\emptyset,\ Y,\ \{a\},\ \{b\},\ \{a,b\}\}.$$

Define $f : X \to Y$ by $f(1) = a,\ f(2) = b,\ f(3) = c$.

Step 1: Find the closed sets.

- *Closed sets in X: complements of τ_X are $\{X,\ \emptyset,\ \{2,3\}\}$.*
- *Closed sets in Y: complements of τ_Y are $\{Y,\ \emptyset,\ \{b,c\},\ \{a,c\},\ \{c\}\}$.*

Step 2: Check each closed set in X.

- *$f(X) = Y$ — closed in Y. ✓*
- *$f(\emptyset) = \emptyset$ — closed in Y. ✓*
- *$f(\{2,3\}) = \{b,c\}$ — is $\{b,c\}$ closed in Y? **Yes.** ✓*

Conclusion: *Interestingly, this map* is *a closed map! Let us modify it to get a failure. Now redefine: $f(1) = a,\ f(2) = a,\ f(3) = b$.*

- *$f(\{2,3\}) = \{a,b\}$. Is $\{a,b\}$ closed in Y? The complement of $\{a,b\}$ is $\{c\}$. Since $\{a,b\} \notin \tau_Y$ as a* complement of something in *τ_Y: we check — τ_Y contains $\{a,b\}$ as an open set, so $\{a,b\}$* is open, *not closed. Hence $f(\{2,3\}) = \{a,b\}$ is open but **not closed**.*

Conclusion (revised f): *This modified f is **not a closed map**.*

Example 3.2 (Constant Map is Always Closed (and Open)). *Let (X, τ_X) and (Y, τ_Y) be any topological spaces. Define $f : X \to Y$ by $f(x) = y_0$ for all $x \in X$, where $y_0 \in Y$ is fixed.*

For any closed set $B \subseteq X$:

$$f(B) = \begin{cases} \emptyset & \text{if } B = \emptyset \\ \{y_0\} & \text{if } B \neq \emptyset. \end{cases}$$

Both $\emptyset$ and $\{y_0\}$ are closed in Y provided *$\{y_0\}$ is a closed set in Y.*

Note: *If Y carries the indiscrete topology $\{\emptyset, Y\}$, then $\{y_0\}$ may* not *be closed unless Y is a singleton. In a T_1 space, however, every singleton is closed, so constant maps into T_1 spaces are always closed maps.*

Example 3.3 (Checking Both Open and Closed Simultaneously). *Let $X = \{p,q\}$, $Y = \{1,2,3\}$, with:*

$$\tau_X = \{\emptyset, X, \{p\}\}, \qquad \tau_Y = \{\emptyset, Y, \{1\}, \{2,3\}\}.$$

Define $f : X \to Y$ by $f(p) = 1,\ f(q) = 2$.

Open Map Check:

Open set in X	*Image*	*Open in Y?*
$\emptyset$	$\emptyset$	*Yes* ✓
X	$\{1,2\}$	*Is $\{1,2\} \in \tau_Y$? No* $\times$
$\{p\}$	$\{1\}$	*Yes* ✓

*$f(X) = \{1,2\} \notin \tau_Y$, so f is **not an open map**.*

Closed Map Check:

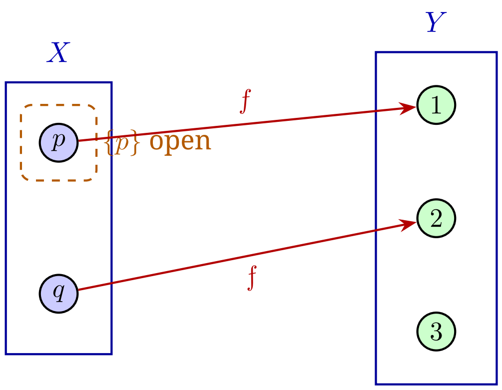
Closed sets in X: $\{X, \emptyset, \{q\}\}$.

Closed sets in Y: complements of τ_Y are $\{Y, \emptyset, \{2,3\}, \{1\}\}$.

Closed set in X	*Image*	*Closed in Y?*
X	$\{1,2\}$	*Is $\{1,2\}$ closed? Its complement is $\{3\}$, not open* $\times$

f *is **not a closed map** either.*

Lesson: A map can be neither open nor closed!



$f(p) = 1,\ f(q) = 2$. The element $3 \in Y$ has no preimage.

Figure 3: Diagram for Example 3.3: f maps $p \mapsto 1$ and $q \mapsto 2$. The image $f(X) = \{1,2\} \subsetneq Y$. Such a non-surjective map can fail to be open or closed because images may fall outside the required open/closed sets of Y.

Common Mistakes: Closed Maps

1. **Forgetting that $\emptyset$ and X are both open *and* closed.** Their images always satisfy both conditions trivially. Do not count these as evidence that a map is open or closed — focus on the non-trivial sets.

2. **Confusing closed map with the inverse of an open map.** A map can be open without being closed and vice versa. Do not assume: "if the map is open, then its inverse is closed." These are unrelated without further assumptions.

Section 3 Summary: Closed Maps

- **Definition:** $f : X \to Y$ is a **closed map** if $f(B)$ is closed in Y for every closed set $B \subseteq X$.

- **Finding closed sets:** Take the complement (relative to the whole space) of each open set.

- **Key check:** Verify the image of every closed set. One failure is enough to disqualify.

- **Both or neither:** A map can be open but not closed, closed but not open, both, or neither.

- **Trivial closed sets:** $\emptyset$ and X are always closed; their images satisfy the condition automatically.

4 Summary

It is important not to confuse open/closed maps with continuous maps. The direction of the mapping and the type of set (image vs. inverse image) are the key differences.

Table 1: Comparison of Continuous, Open, and Closed Maps

Map Type	**Definition**	**Direction of Check**
Continuous Map	The **inverse image** of every **open** set in Y is **open** in X.	$Y \to X$ (Backward)
Open Map	The **image** of every **open** set in X is **open** in Y.	$X \to Y$ (Forward)
Closed Map	The **image** of every **closed** set in X is **closed** in Y.	$X \to Y$ (Forward)

A map can be open, closed, continuous, any combination of these, or none of them.

4.1 Visual Summary: The Four Combinations

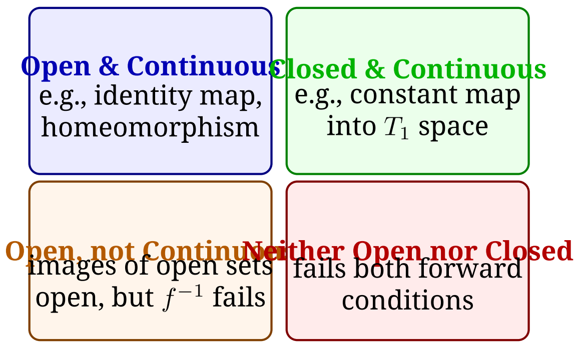


Figure 4: The four possible combinations of open/closed map properties. Note: "continuous" is an additional independent axis not shown here.

4.2 A Complete Worked Example: Testing All Three Properties

Full Analysis Example

Let $X = \{1,2,3\}$, $Y = \{a,b,c\}$, with:

$$\tau_X = \{\emptyset, X, \{1\}, \{2\}, \{1,2\}\}, \qquad \tau_Y = \{\emptyset, Y, \{a\}, \{a,b\}\}.$$

Define $f : X \to Y$ by $f(1) = a,\ f(2) = b,\ f(3) = c$.
Closed sets:

- In X: $\{X, \emptyset, \{2,3\}, \{1,3\}, \{3\}\}$
- In Y: $\{Y, \emptyset, \{b,c\}, \{c\}\}$

(a) Is f continuous? Check $f^{-1}(V)$ for each open $V \in \tau_Y$:

- $f^{-1}(\emptyset) = \emptyset \in \tau_X$. ✓
- $f^{-1}(Y) = X \in \tau_X$. ✓
- $f^{-1}(\{a\}) = \{1\} \in \tau_X$. ✓
- $f^{-1}(\{a,b\}) = \{1,2\} \in \tau_X$. ✓

f is **continuous.** ✓
(b) Is f an open map? Check $f(U)$ for each open $U \in \tau_X$:

- $f(\emptyset) = \emptyset \in \tau_Y$. ✓
- $f(X) = Y \in \tau_Y$. ✓
- $f(\{1\}) = \{a\} \in \tau_Y$. ✓
- $f(\{2\}) = \{b\}$. Is $\{b\} \in \tau_Y$? **No!** τ_Y has $\{a\}$ and $\{a,b\}$ but not $\{b\}$. $\times$

f is **not an open map**.
(c) Is f a closed map? Check $f(B)$ for each closed B in X:

- $f(X) = Y$ — closed. ✓
- $f(\emptyset) = \emptyset$ — closed. ✓
- $f(\{2,3\}) = \{b,c\}$ — closed in Y. ✓
- $f(\{1,3\}) = \{a,c\}$ — is $\{a,c\}$ closed? Its complement $\{b\}$ must be open. But $\{b\} \notin \tau_Y$. So $\{a,c\}$ is **not closed**. $\times$

f is **not a closed map**.
Final Answer: f is *continuous*, but *neither open nor closed.*

5 Practice Problems

Let $X = \{a,b\}$ and $Y = \{p,q,r\}$. Consider the map $f : X \to Y$ defined by $f(a) = p$ and $f(b) = r$. Determine if f is an open map and/or a closed map for each of the following topologies.

1. $\tau_X = \{\emptyset, X, \{a\}\}$
   $\tau_Y = \{\emptyset, Y, \{p\}\}$

2. $\tau_X = \{\emptyset, X, \{b\}\}$
   $\tau_Y = \{\emptyset, Y, \{p, q\}, \{r\}\}$

3. $\tau_X = \{\emptyset, X, \{a\}, \{b\}\}$ (Discrete Topology)
   $\tau_Y = \{\emptyset, Y, \{p, r\}\}$

Hints for Practice Problems

- **Problem 1:** There are 3 open sets in X. Find f applied to each. Then repeat for closed sets.
- **Problem 2:** Note that $f(b) = r$ and $\{r\}$ is open in Y — does this help or hurt?
- **Problem 3:** With the discrete topology on X, *every* subset is open (and closed). You must check *all four* subsets of $X = \{a, b\}$: $\emptyset, \{a\}, \{b\}, X$.

Overall Lecture Summary

- **Open Map:** $f : X \to Y$ sends every open set in X to an open set in Y (forward direction).
- **Closed Map:** $f : X \to Y$ sends every closed set in X to a closed set in Y (forward direction).
- **Continuous Map:** f^{-1} sends every open set in Y to an open set in X (backward direction).
- **Key formula:** Closed sets of (X, τ) are exactly the complements $\{X \setminus U : U \in \tau\}$.
- **Independence:** These three properties are mutually independent. Always check each one separately.
- **Homeomorphism shortcut:** Bijection + Continuous + Open $\Rightarrow$ Homeomorphism.